Game Theory in Electricity Markets

Week 1 – Warm-up

Why do we have electricity markets?

1880s **Centralizing power supply**

Edison's Pearl Street Station (1882) sells electricity as a commodity. That's cheaper than every building running its own generator and safer than lighting your house with gas.

1880s–1890s **War of the currents**

AC transmission wins the war of currents against DC, as it was easier to transport. Many DC girds become one AC grid.

1930s–1960s **Utilities interconnect into grids**

For reliability and scale, it made sense to connect the smaller grids into national-wide grids. Additionally, it became political will to electrify more and more households.

1960s–2000s **Market structure and reliability upgrades**

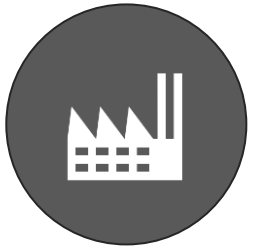
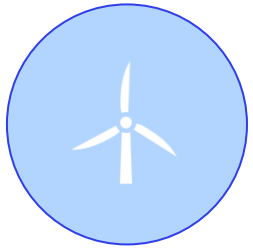
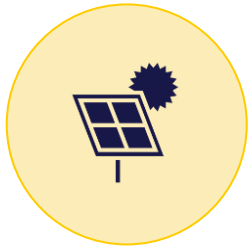
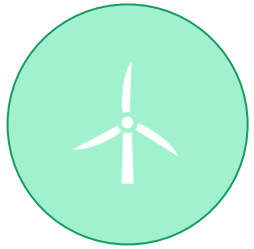
Some say oil shocks, cost overruns, and blackouts expose monopoly inefficiency. Other say it was just the sign of the time. Anyways many countries unbundle generation from the wires to break these monopoles.

2000s–today **Using smart technologies and incorporating renewables**

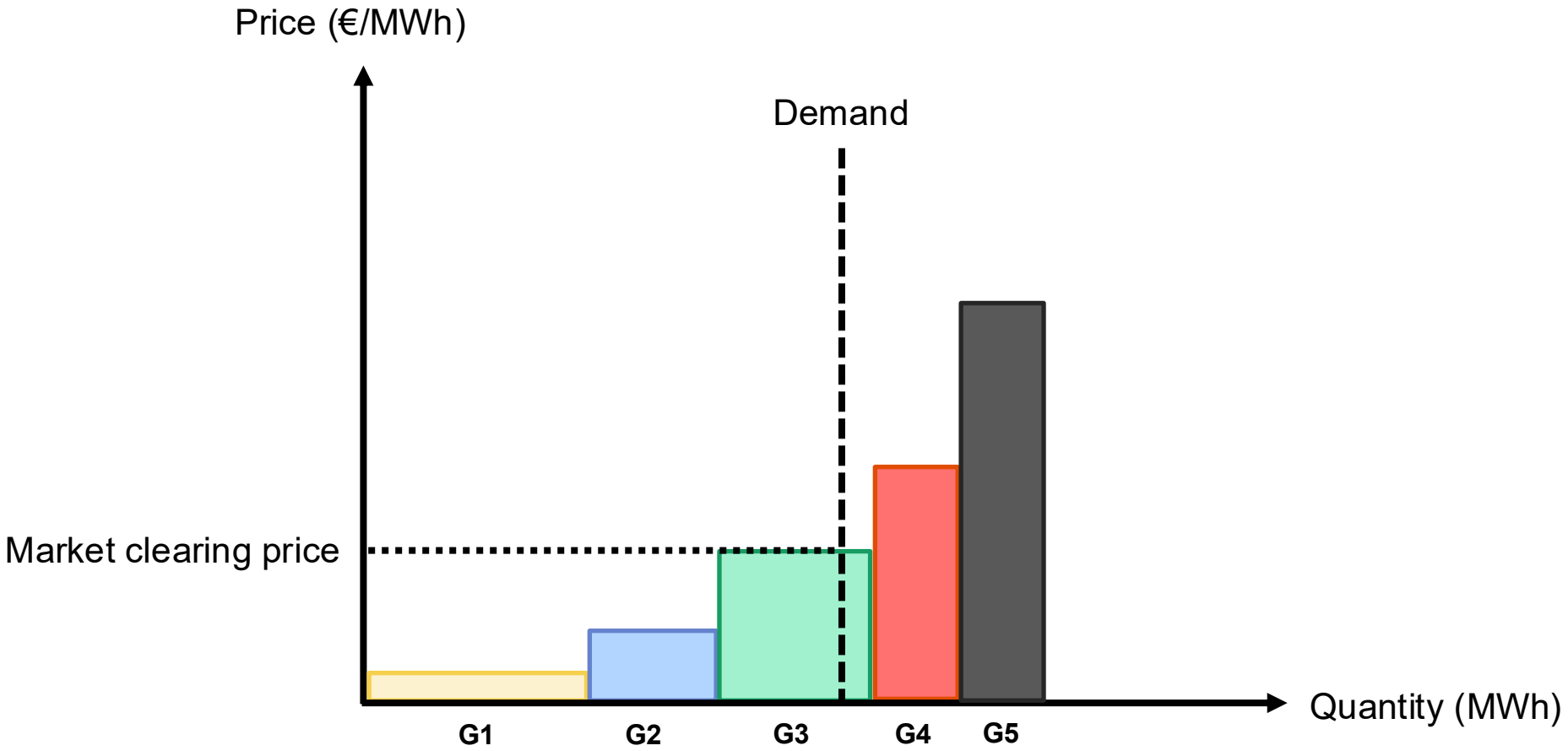
Adjusting the grid to keep it resilient while demand rises. Reacting to the structural changes that come with the incorporation of renewables

How to fulfill the demand in the best way?

We just ask every generator to tell us how much it costed them to produce.

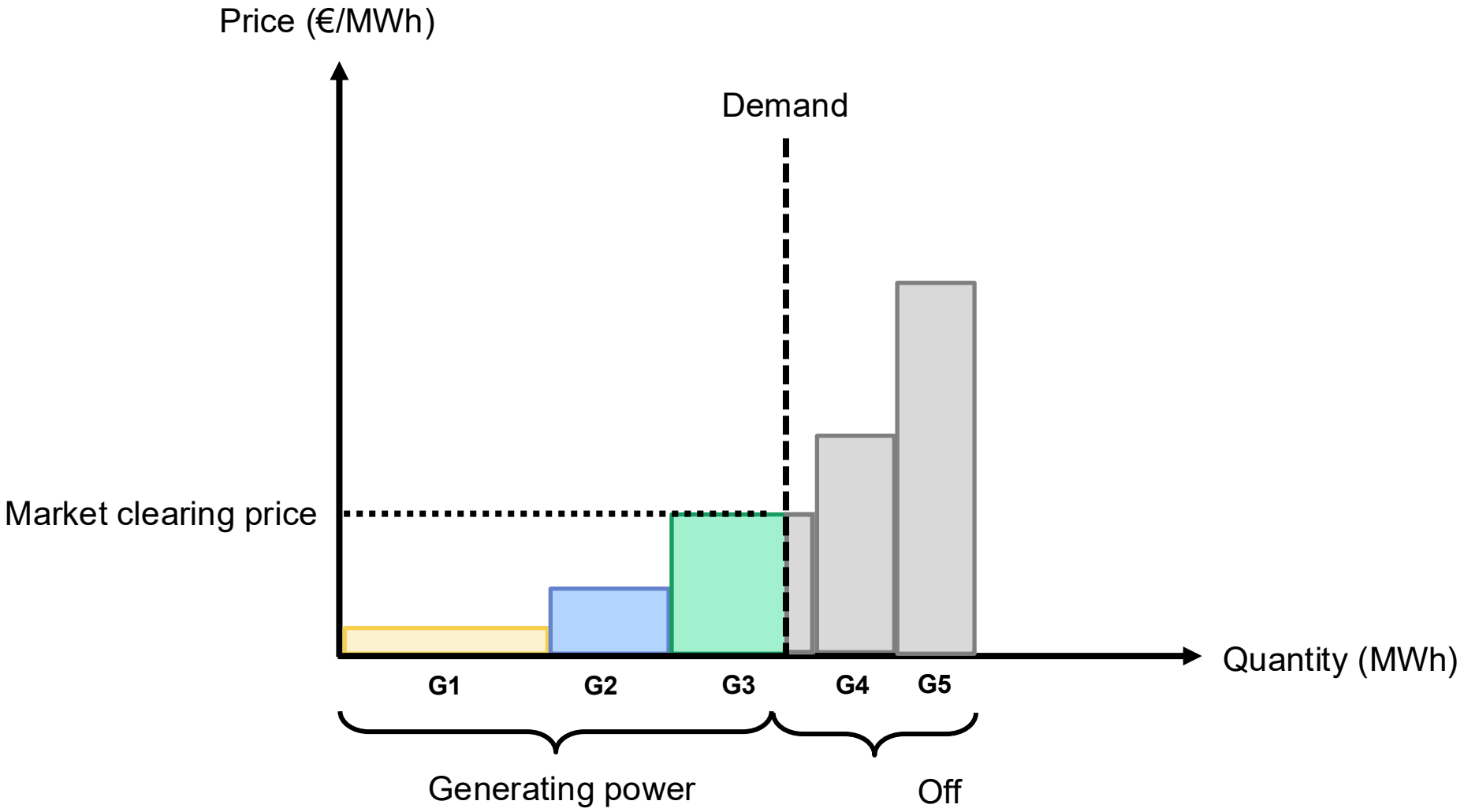
Coal	Gas	Offshore	Solar	Onshore
				
30 MWh at 65$/MWh	30 MWh at 40$/MWh	40 MWh at 12.5$/MWh	100 MWh at 5$/MWh	50 MWh at 25$/MWh

Market Clearing 101

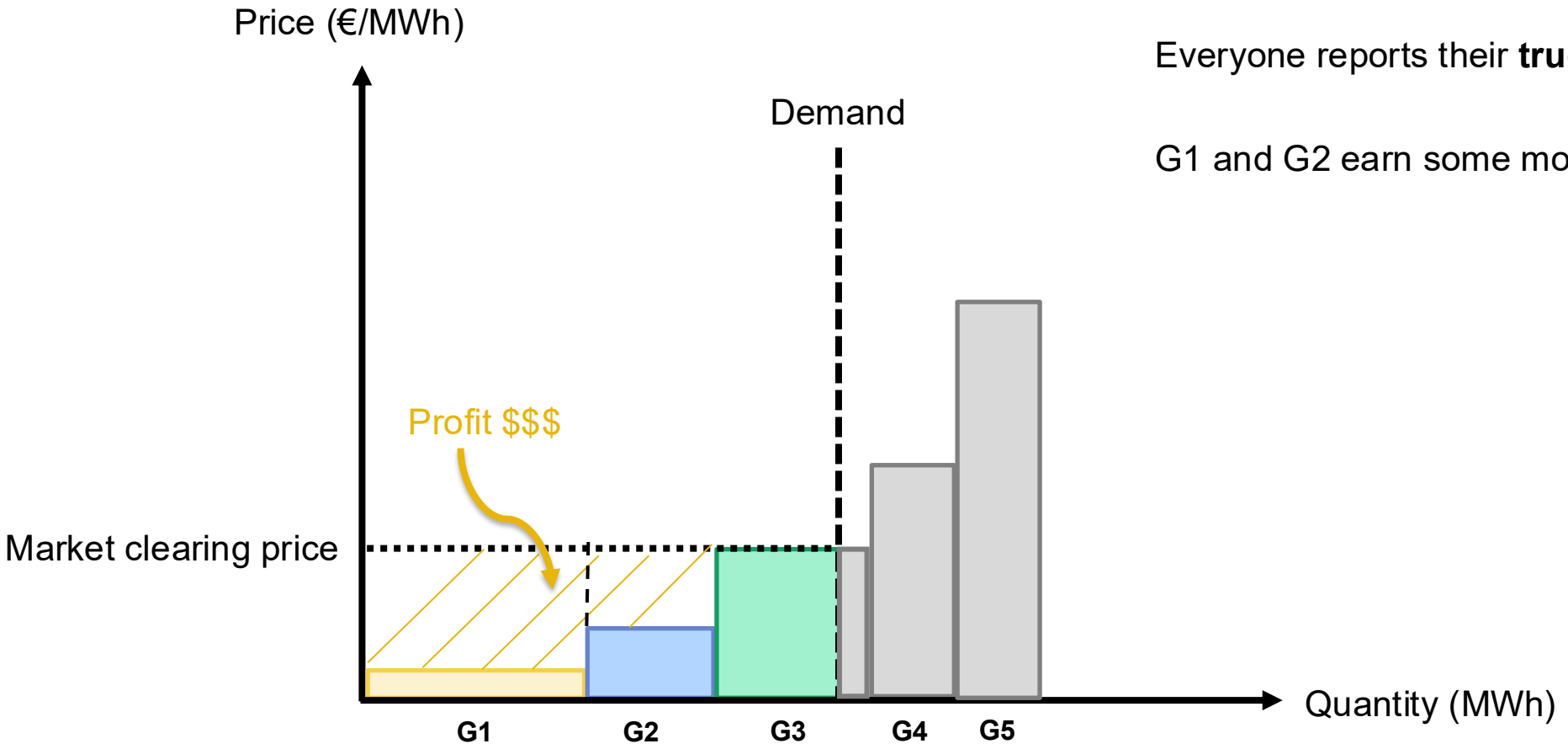


Graphic from: https://www.britannica.com/

Market Clearing 101



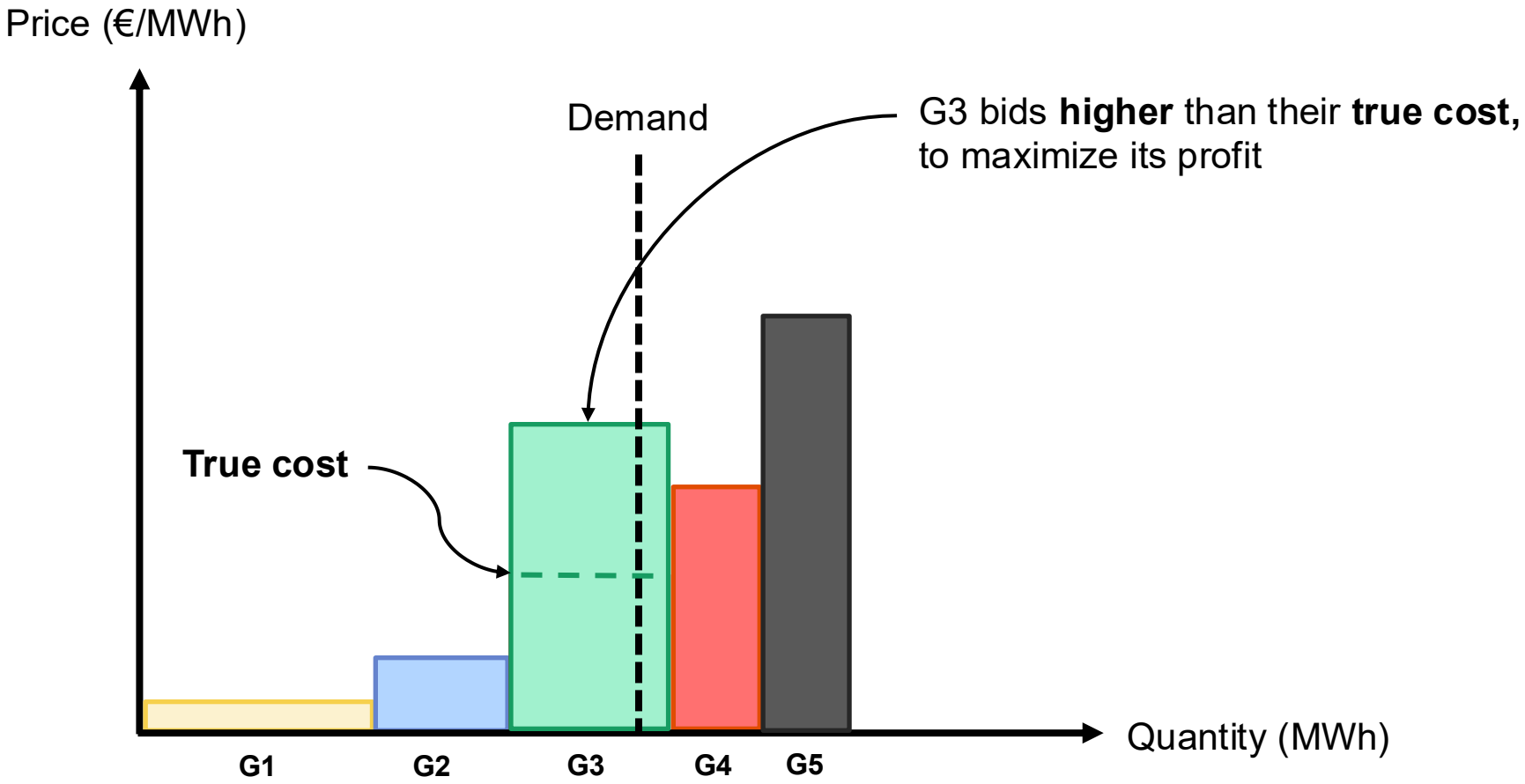
Market Clearing 101



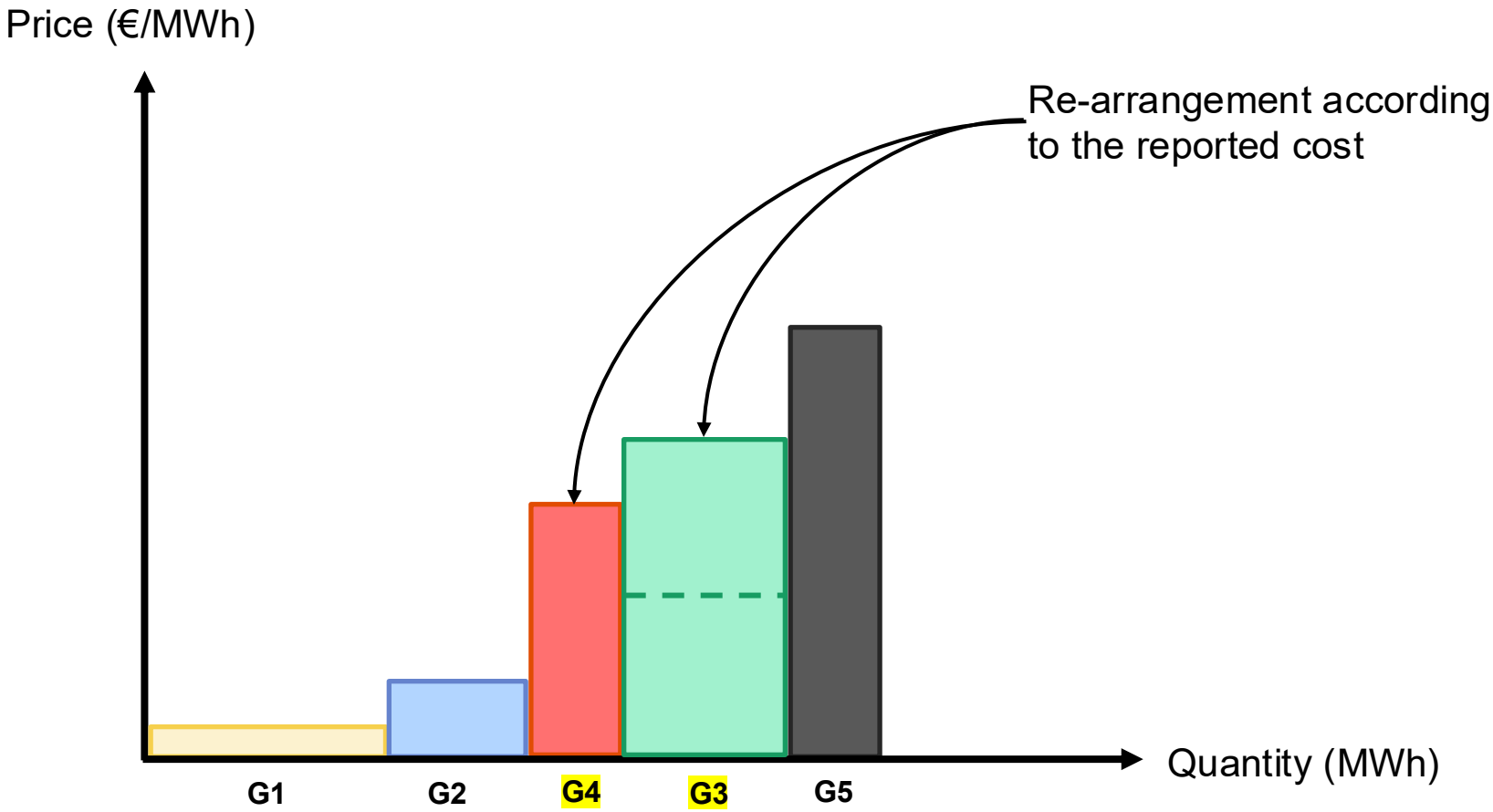
Everyone reports their **true cost**

G1 and G2 earn some money, the rest does not

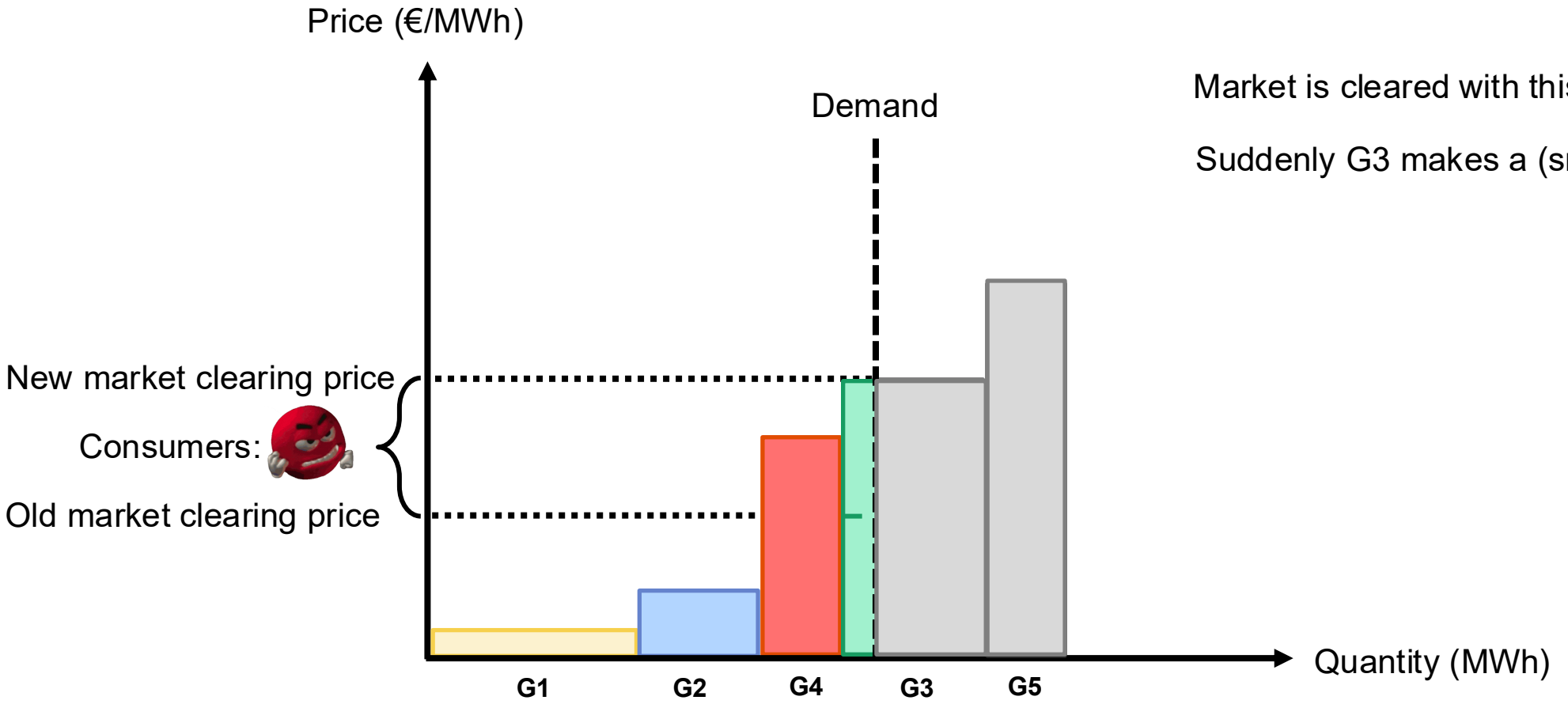
Strategic behaviour



Strategic behaviour



Strategic behaviour

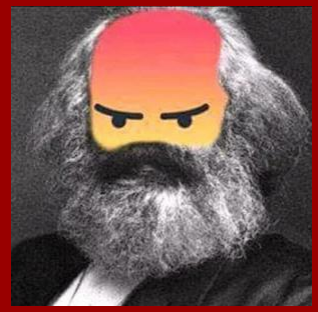


Market is cleared with this information

Suddenly G3 makes a (small) profit



Stop thinking about your money – think of our money!



High prices are not the problem

Consumers pay more = generators earn more



pay more



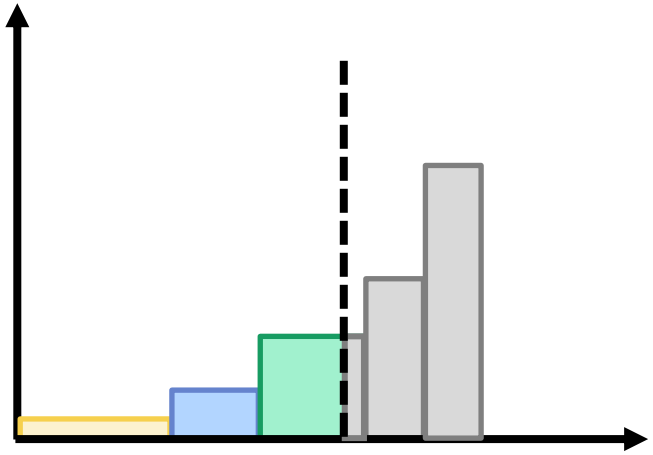
Consumers **Generators**

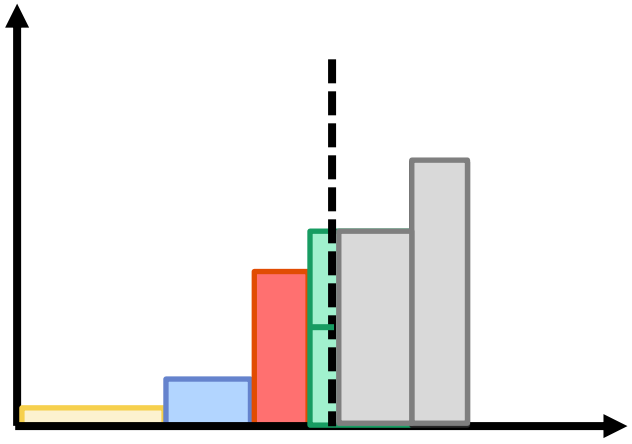
For society the price does not matter

Imag from: https://www.lustiges-taschenbuch.de/

The real problem

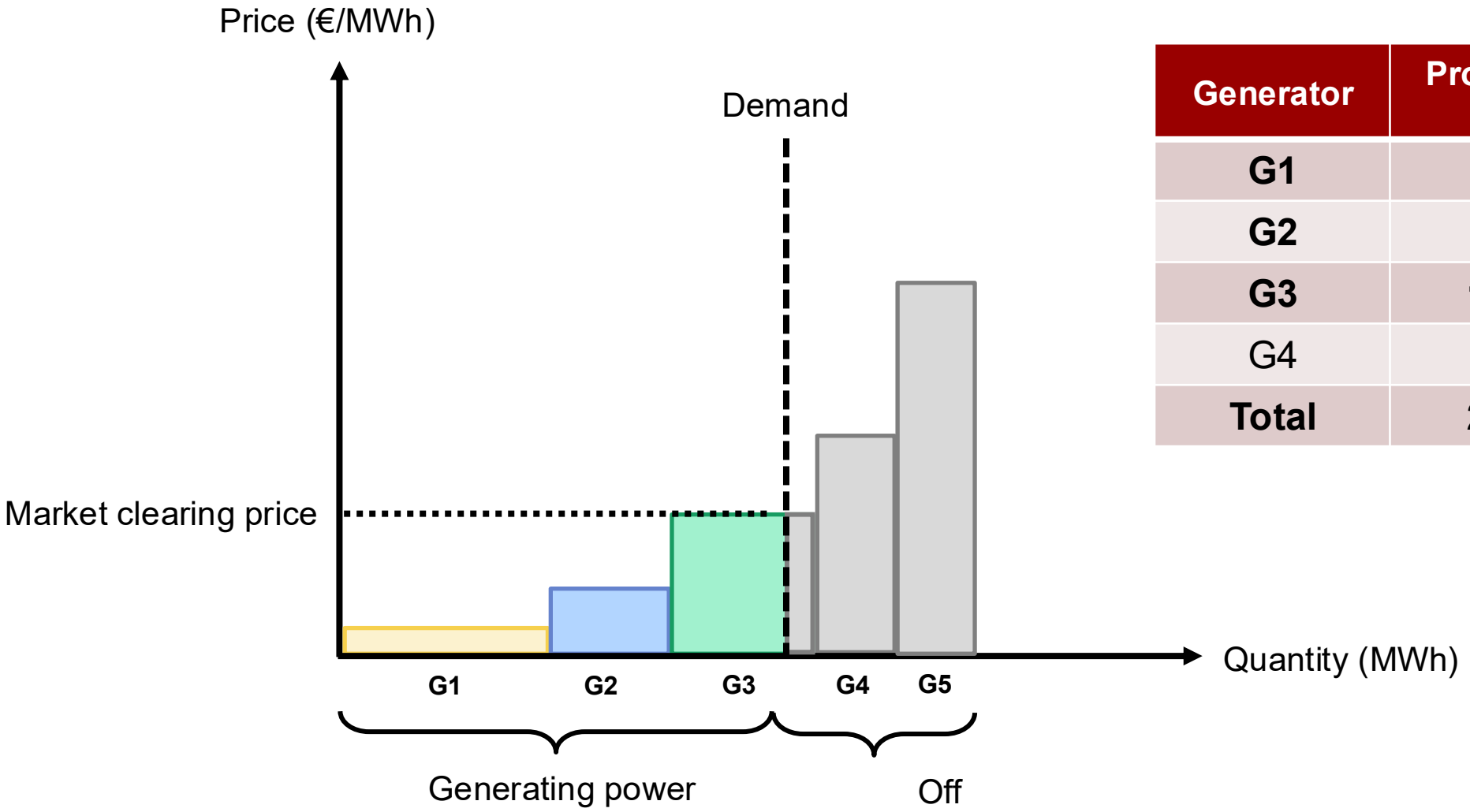
Coal	Gas	Offshore	Solar	Onshore
30 MWh at 55$/MWh	30 MWh at 40$/MWh	40 MWh at 12.5$/MWh	100 MWh at 5$/MWh	50 MWh at 25$/MWh





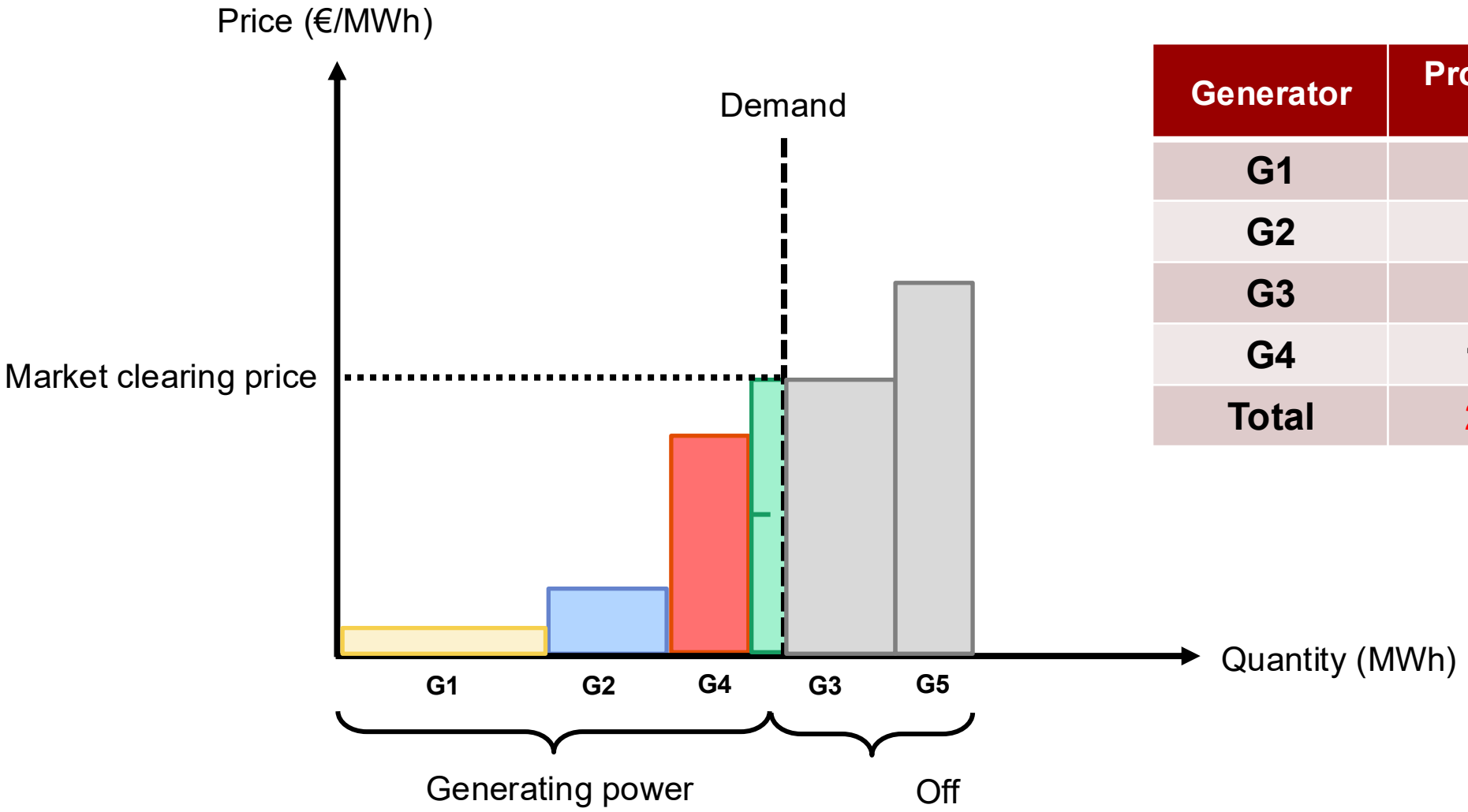
We might use fossil fuels although we do not need to!

Let's formalize that

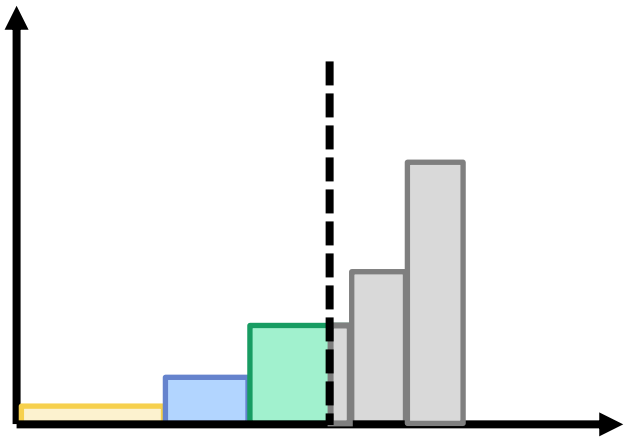


Generator	Producing cost	Generation
G1	**500**	**100%**
G2	**500**	**100%**
G3	**1000**	**80%**
G4	0	0%
Total	**2000**	**-**

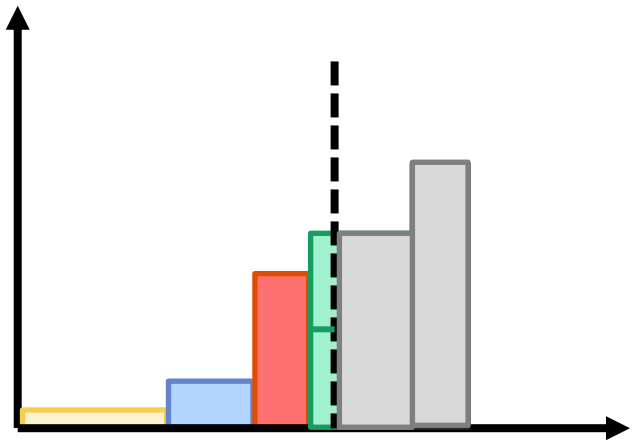
Let's formalize that



Generator	Producing cost	Generation
G1	500	100%
G2	500	100%
G3	250	20%
G4	1200	100%
Total	2450	-



Generator	Producing cost	Generation
G1	500	100%
G2	500	100%
G3	1000	80%
G4	0	0%
Total	2000	-



Generator	Producing cost	Generation
G1	500	100%
G2	500	100%
G3	250	20%
G4	1200	100%
Total	2450	-

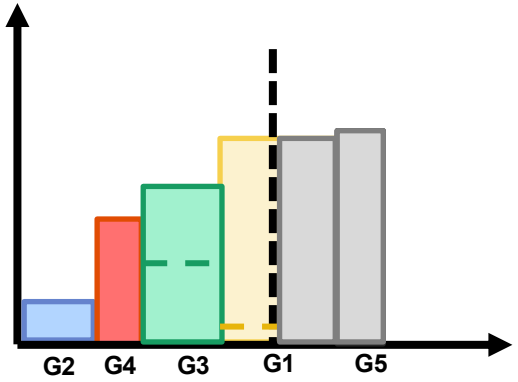
$$Cost\ ratio = \frac{2450}{2000} = 1.225$$

$\longrightarrow$ *The cost is* 22.5% *higher than needed*

Which scenario do you think is more realistic?

It rationally makes sense for G3
to artificially inflate their price

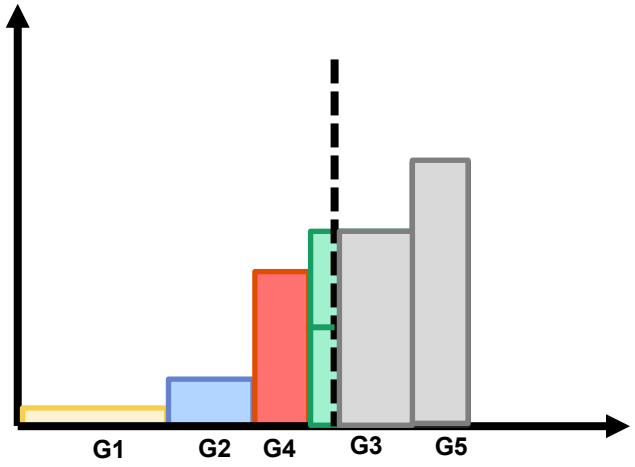
As it could make sense for G1 to do so…



Generator	Producing cost	Generation
G1	300	600%
G2	500	100%
G3	1250	100%
G4	1200	100%
Total	3250	-

$$Cost\ ratio = \frac{3250}{2000} = 1.625,$$
62.5% higher than needed!

Who knows, maybe it can get even worse…



Generator	Producing cost	Generation
G1	500	100%
G2	500	100%
G3	250	20%
G4	1200	100%
Total	2450	-

How to measure inefficiency of a market

"when each participant tries to maximize their profit"

There might be many possible outcomes

$$\textit{Price of Anarchy} = \max \frac{\textit{Cost of the emergent market outcome}}{\textit{Minimum cost}}$$

Price of Anarchy = 1, society pays **as little as possible** to satisfy demand

Price of Anarchy = 2, we might pay **twice as much** to satisfy (same) demand

Should we be angry?

In case you said we should:

- It is rational for them to act like that.
- Thay could occur loses when others act strategically but they do not.
- In some countries it might even be illegal to not do what is best for the company.
- That is just how our world works.

In case you said we shouldn’t:

- We are losing money as a society. That means we are investing work and resources without getting anything out of it.
- We should not burn fossil fuels if we do not need to do so.
- In some systems it is illegal to lie about your producing costs.

Let’s see how others handle such situation

- https://www.youtube.com/watch?v=hdK4vPz0qaI&t=10s

- One can also watch:
  - https://www.youtube.com/watch?v=TbuD-6BbnQw&t=184s
  - https://www.youtube.com/watch?v=M0mHp3iyWGg

Losing on purpose: London 2012

The setup London 2012 introduced a round-robin group stage before the knockouts, the first time in Olympic badminton history.

The incentive A shock upset reshuffled the bracket. Several top pairs realized that winning their last group match could mean a tougher tournament tree than losing it. (All because a Danish team performed better than expected.)

What happened Four pairs served into the net and hit shots long, in front of a booing crowd, until the referee stopped the matches.

The ruling The Badminton World Federation disqualified all eight players for “not using one’s best efforts to win.”

Nobody broke a rule. The rules were the problem.

The Disgrace of Gijón: 1982

The setup 1982 World Cup, final group match: West Germany vs Austria. It is played a day after Algeria had already finished their own group games.

The incentive West Germany and Austria both knew Algeria's results in advance: a West German win by one or two goals would send both of them through on goal difference, eliminating Algeria.

What happened Horst Hrubesch scored for West Germany in the 10th minute. For the remaining 80 minutes, both teams just passed the ball around, protecting the exact scoreline they both needed. Fans chanted “Que se besen!” (“Let them kiss!”). A German commentator refused to keep commentating.

The ruling No rule was broken, so no one was disqualified. The FIFA rewrote the rules: from 1986 onward, every group's final two matches kick off at the same time.

This time nobody was punished the FIFA changed the rules instead.

Why we need mechanism design

The pattern In Gijón and in London, skilled, rational people made the strategically optimal choice and it made everyone worse off.

The parallel An electricity market is the same kind of game. Strategic bidding pushed up total cost significantly. This extra cost is payed by everyone on the grid.

The stakes A badminton match is one bad afternoon. A market clears every settlement period (15 min), forever, and the bill goes to households and industry. That's worse than a stadium of booing fans.

The task Mechanism design asks the Gijón question before the game, not after: how do we design the rules of the market so that self-interested bidding still gives low Price of Anarchy?

We can't fix the players. We can design the game.

Why is this so hard?

Price of Anarchy

Level 3 — Maximize the cost ratio over all such settings

Level 2 — Players maximize revenue

Level 1 — The operator clears the market, given fixed bids

The market clearing in itself is an optimization problem!

Mechanism Design

Level 4 — Minimize over the space of rules

Price of Anarchy — Maximize the cost ratio over all such settings

Level 2 — Players maximize revenue

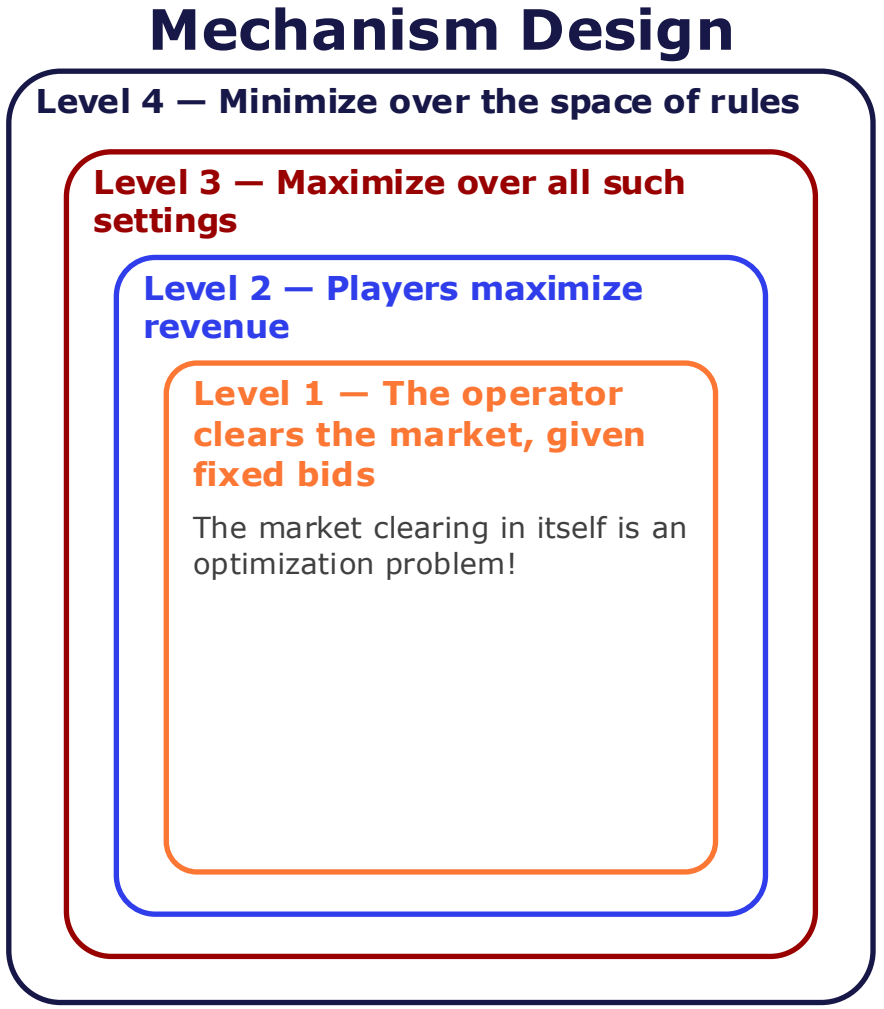
Level 1 — The operator clears the market, given fixed bids

The market clearing in itself is an optimization problem!

Generally, already level 1 + 2 is computationally hard and needs additional assumptions, for example on how much players know about each other.

Nevertheless, our goal and your final project will be to use mechanism design to make qualitative statements on how to lower the inefficiency in electricity markets.

This is an ongoing research topic. If you get interesting results, we may take them further in a master thesis and/or publish them together.



Plan for the next weeks



The project phase continues for the rest of the semester.

The project

- You work in groups of 3
- Your grade only depends on the quality and correctness of this report
- With every group together I define the task of the report
- Use AI as much as you want
- The report needs to be short; maximum 8 standard latex pages
- The idea is that you extend the result section of a paper by me to get further insights
- The paper is “only” on how to bound PoA. Projects could be:
  - Use the method on different models for electricity markets
  - Use the method within mechanism design
  - Asses the quality of the bound

Literature

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- Roughgarden, T. (2013). *Lecture Notes on Algorithmic Game Theory* (Stanford CS364A). https://theory.stanford.edu/~tim/f13/f13.pdf
- Solé Àvila, M., Ebrahimy, R., & Tsaousoglou, G. (2025). *How Inefficient Can an Electricity Market Be?* In Proceedings of the 2025 IEEE Kiel PowerTech (pp. 1–6). IEEE.
- Strauch, L, Kamgarpour, M. & Tsaousoglou, G. (2026, not published yet). *Electricity Markets as Games: Bounding the Price of Anarchy*


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